

Ming Zhong

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SUMMARY

Self-motivated Senior Data Scientist with 5+ years of experience in the retail industry. Expertise leveraging analytical techniques to transform data into business intelligence. Proven track record of collaborating across teams in agile framework, developing ETL pipelines, and providing data-driven solutions to drive business growth and operational efficiency.

SKILLS

Languages/Tools: Python, SQL, R, Spark, Java, JavaScript, HTML, Tableau, PowerBI

Frameworks: Agile, Pyspark, TensorFlow, MLflow, Pyomo, Keras, Scikit-learn, Pandas, PyTorch, nloptr

Platforms: Databricks, AWS (certified solution architect), GCP, Azure, Git

PROFESSIONAL EXPERIENCE

Senior Data Scientist, Circle K Stores Inc., Tempe, AZ

Aug. 2022 - Present

- **Analog Site Tool:** Identified comparable stores for financial forecasting using Mahalanobis distance and ML-driven feature selection, enabling data-driven site analysis for Real Estate teams.
- **White Space Tools:** Built and deployed random forest models to predict ROI in 220K+ block groups, identifying 15K+ high-potential locations and narrowing real estate planning efforts in US and Canadian markets.
- **Site Selection Model:** Enhanced revenue prediction accuracy by 15% through the development of a random forest regression model, preventing potential real estate investment failures exceeding \$2M in the Texas sector.
- **Churn Prediction:** Engineered an end-to-end ML pipeline (Scikit-Learn, MLflow), achieving > 0.9 accuracy and > 0.82 AUC to predict loyalty program churn; Conducted customer segment analysis, visualizing key drivers to guide retention strategies.
- **Pricing Optimization:** Developed a multi-objective pricing optimizer using genetic algorithms and compromise programming, reducing runtime 10x while delivering 1–3% annual profit uplift through rule-based strategic pricing.

Data Scientist, Circle K Stores Inc., Tempe, AZ

June 2021 - July 2022

- **Carwash Demand Forecast:** Predicting carwash demand using various forecasting models like Prophet, ARIMA, RNNs, and LSTM enabled stakeholders to optimize supply chain and pricing, resulting in cost savings exceeding \$500k.
- **Localized Pricing Engine:** Led migration of pricing solution from Databricks to Azure Data Factory, reducing pricing refresh cycle time by 88% (70 hours) and cutting costs by 76% (\$500).
- **Localized Pricing:** Estimated price elasticity for each SKU using Unobserved Component Model (UCM) for time series forecasting to optimize product pricing strategies, resulting in a 3% average margin lift (over \$120M in gross margin) annually.
- **Cigarettes A/B Testing:** Designed and implemented A/B testing to measure the impact of price changes for Cigarettes in test stores which resulted in a margin uplift of \$700,000 across the Midwest region.
- **ETL Pipeline:** Engineered data pipelines to track the state of the storage and ensure data quality for modeling and business purposes, eliminating over 60 hours of manual labor per quarter.

Business Analyst, ROKT Corp., New York, NY

Jan. 2020 - May 2021

- **Conversions Analytics:** Evaluated attribution identifier performance using SQL and recommended including phone numbers in the attribution model, driving a 40% increase in ad conversions.
- **COVID Dashboard:** Created Tableau COVID Recovery Dashboards tracking more than 10M transactions for 10+ major clients, providing critical insights for relaunch decisions for ad campaigns.
- **Account Managers ChatBot:** Developed an NLP-powered chatbot in Google Chat using Python and NLP libraries (spaCy, NLTK) to answer client campaign performance questions, winning first place in the company's hackathon.

Data Science Intern, Roche Holding AG, New York, NY

Jul. 2019 - Aug. 2019

- **R Library Development:** Developed an R package with data cleaning functions, cohort balancing using propensity scores, multiple hypothesis testing, and visualization for real-world evidence (RWE) data, which is used by 3+ researchers.
- **Patient Journey Dashboard:** Enhanced, deployed and maintained an R Shiny App to visualize multidimensional patient time series data to help gain clinical insight, making it easy to share clinical result among teams

EDUCATION

Georgia Institute of Technology

Master of Science, Computer Science — GPA: 4.0

Atlanta, GA

Aug. 2023 - Present

Columbia University

Master of Arts, Statistics — GPA: 3.6

New York, NY

Dec. 2019

University of Macau

Bachelor of Science, Finance — GPA: 3.7

Macau, China

Jul. 2018